AWS VPC Peering Project

Connecting two isolated Amazon VPCs so that resources in each can talk to each other over private IP addresses, using a bastion host as the entry point.

- **Region:** Asia Pacific (Sydney) — `ap-southeast-2`
- **Tools:** AWS Management Console (VPC, EC2), OpenSSH
- **Report:** see `Project-Report.pdf`

Architecture

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Component	VPC A	VPC B
VPC name / ID	`ewb-vpc-a` / `vpc-014c70892d7a0e4f8`	`ewb-vpc-b` / `vpc-080c5958b2352dac7`
CIDR	`10.10.0.0/16`	`10.20.0.0/16`
Subnet	`ewb-a-public` (ap-southeast-2a)	`ewb-b-private`
Route table	`ewb-rt-a-public` (`0.0.0.0/0` → IGW)	`ewb-rt-b-private` (local only)
Internet gateway	`ewb-igw-a`	none (private)
Security group	`ewb-sg-bastion`	`ewb-sg-b`
Compute	EC2 `ewb-bastion` (t3.micro, `10.10.1.161`)	private EC2 (target)

Peering connection: `ewb-a-b-peer` (`pcx-03178b55064bce331`), requester VPC A → accepter VPC B, same account and region.

> The two CIDR blocks do not overlap (`10.10.0.0/16` vs `10.20.0.0/16`), which is a hard requirement for VPC peering.

What was built

1. Created `ewb-vpc-a` (`10.10.0.0/16`) and `ewb-vpc-b` (`10.20.0.0/16`).
2. Created the public subnet `ewb-a-public` in VPC A and the private subnet `ewb-b-private` in VPC B.
3. Created and attached internet gateway `ewb-igw-a` .
4. Created route table `ewb-rt-a-public` , added `0.0.0.0/0 → igw-056cb8e2394cefccb` , and associated it with the public subnet.
5. Created route table `ewb-rt-b-private` and associated it with the private subnet.
6. Created security groups `ewb-sg-bastion` (SSH access) and `ewb-sg-b` , and edited inbound rules.

7. Launched the `ewb-bastion` EC2 instance and connected with SSH key authentication (`ewb-lab.pem`).
8. Requested the VPC peering connection `ewb-a-b-peer` (status: *Pending acceptance*).

Steps still required to finish the peering test

The screenshots stop at the pending peering request. To complete and verify connectivity:

1. **Accept** the peering request (Peering connections → Actions → Accept request).
2. **Route tables:** in `ewb-rt-a-public` add `10.20.0.0/16 → pcx-03178b55064bce331`; in `ewb-rt-b-private` add `10.10.0.0/16 → pcx-03178b55064bce331`.
3. **Security group:** in the security group attached to the private instance, allow SSH (22) and ICMP from `10.10.0.0/16`.
4. **Test** from the bastion: `ping <private-ip-of-VPC-B-instance>` and `ssh ec2-user@<private-ip>`.
5. *(Optional)* Enable DNS resolution over the peering connection.

Issue to double-check

In the Security Groups list, `ewb-sg-b` is shown under `vpc-0395efd1798c35bb0`, which is **not** `ewb-vpc-b` (`vpc-080c5958b2352dac7`). A security group can only be attached to instances in its own VPC. If `ewb-sg-b` is meant for VPC B, recreate it inside `ewb-vpc-b`.

Repository layout

```
AWS-VPC-Peering-Project/
├── README.md              project explanation
├── Project-Report.pdf     full project report
├── Screenshots/           AWS console screenshots (01–13)
└── Architecture/          architecture diagram
```

Key concepts

- **VPC peering** is a one-to-one, non-transitive network link between two VPCs using private IPv4/IPv6 addresses; no gateway, VPN or separate hardware is needed.
- **Routing is manual:** both sides need a route to the other VPC's CIDR via the peering connection.
- **Security groups still apply:** peering does not open any ports by itself.
- **Bastion host:** the only instance with a public entry point; private instances are reached through it.